Development of a Polyamine Database for Assessing Dietary Intake

Reducing the concentration of polyamines (spermine, spermidine, and putrescine) in the body pool may slow the cancer process. Because dietary spermine, spermidine, and putrescine contribute to the body pool of polyamines, quantifying them in the diet is important. Limited information about polyamine content of food is available, especially for diets in the United States. This brief report describes the development of a polyamine database linked to the Fred Hutchinson Cancer Center food frequency questionnaire (FFQ). Values for spermine, spermidine, and putrescine were calculated and reported per serving size (nmol/serving). Of the foods from the database that were evaluated, fresh and frozen corn contain the highest levels of putrescine (560,000 nmol/serving and 902,880 nmol/serving) and spermidine (137,682 nmol/serving and 221,111 nmol/serving), and green pea soup contains the highest concentration of spermine (36,988 nmol/serving). The polyamine database and FFQ were tested with a convenience sample (n=165). Average daily polyamine intakes from the sample were: 159,133 nmol/day putrescine, 54,697 nmol/day spermidine, and 35,698 nmol/day spermine. Orange and grapefruit juices contributed the greatest amount of putrescine (44,441 nmol/day) to the diet. Green peas contributed the greatest amount of spermidine (3,283 nmol/day) and ground meat contributed the greatest amount of spermine (2,186 nmol/day). Development of this database linked to an FFQ provides a means of estimating polyamine intake and contributes to investigations relating polyamines to cancer.